

Image Classification Extension

<https://github.com/aiyshwary/Image-Classification-Extension-in-Rapid-Miner>

OVERVIEW

A RapidMiner Studio extension enabling no-code transfer learning for image classification, supporting various architectures on custom datasets and ONNX-based inference without writing a single line of code.

IMPACT

Developed a RapidMiner extension with two custom operators — Fine Tuning and Inference — enabling no-code transfer learning with ResNet, VGGNet, DenseNet, Inception and more using a drag-and-drop interface.

KEY DETAILS

1. Fine Tuning operator accepts an image directory and hyperparameters; outputs a fine-tuned model.
2. Inference operator loads the ONNX model and outputs a prediction DataFrame with classified images.
3. Supports ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, and Inception architectures.
4. Compatible with both CPU and GPU (CUDA) environments via configurable conda environment.
5. Installed as a JAR extension in RapidMiner Studio with Python Scripting backend.
6. Inference operator requires four inputs: the ONNX model path, image path, class-label text file, and classified output images alongside the prediction DataFrame.
7. Depends on Custom Operators ($\geq 1.1.0$) and Python Scripting ($\geq 10.0.1$) marketplace extensions in a compatible environment (Python 3.10.8, ONNX 1.15.0, ONNX Runtime 1.15.1).